

2.2 DC and AC Simulations

A DC sweep of V_{ID} from -0.5 V to 0.5 V was performed in order to analyze the response of the circuit under differential excitation. This made it possible to plot the differential output voltage V_{OD} and the common-mode output voltage V_{OC} as functions of V_{ID} . In addition, the drain currents of the two transistors forming the differential pair were plotted versus V_{ID} . From the operating-point list of the transistors, several important quantities can also be extracted, including the drain current I_D , the drain-source voltage V_{DS} , the transconductance g_m , the intrinsic small-signal output resistance r_o , and the overdrive voltage V_{OV} .

Table 2: Transistor operating-point parameters

	$ V_{OV} $ [mV]	$ V_{DS} $ [mV]	$ I_D $ [μ A]	g_m [mS]	r_o [k Ω]
M1/M2	185.998	455.619	118.02	1.082	8.626
M3/M4	145.017	461.178	118.02	1.237	25.501
M5/M6	145.238	600.749	616.402	6.396	5.471
M7/M8	171.213	599.251	616.388	6.151	5.663
M9	172.629	283.203	236.04	2.327	10.202
M10	174.918	456.554	50	0.4904	67.442

From Fig. 17, it can be observed that the common-mode output voltage remains approximately constant at 600 mV. The differential output voltage, instead, is very small at $V_{ID} = 0$, as expected for a balanced differential input.

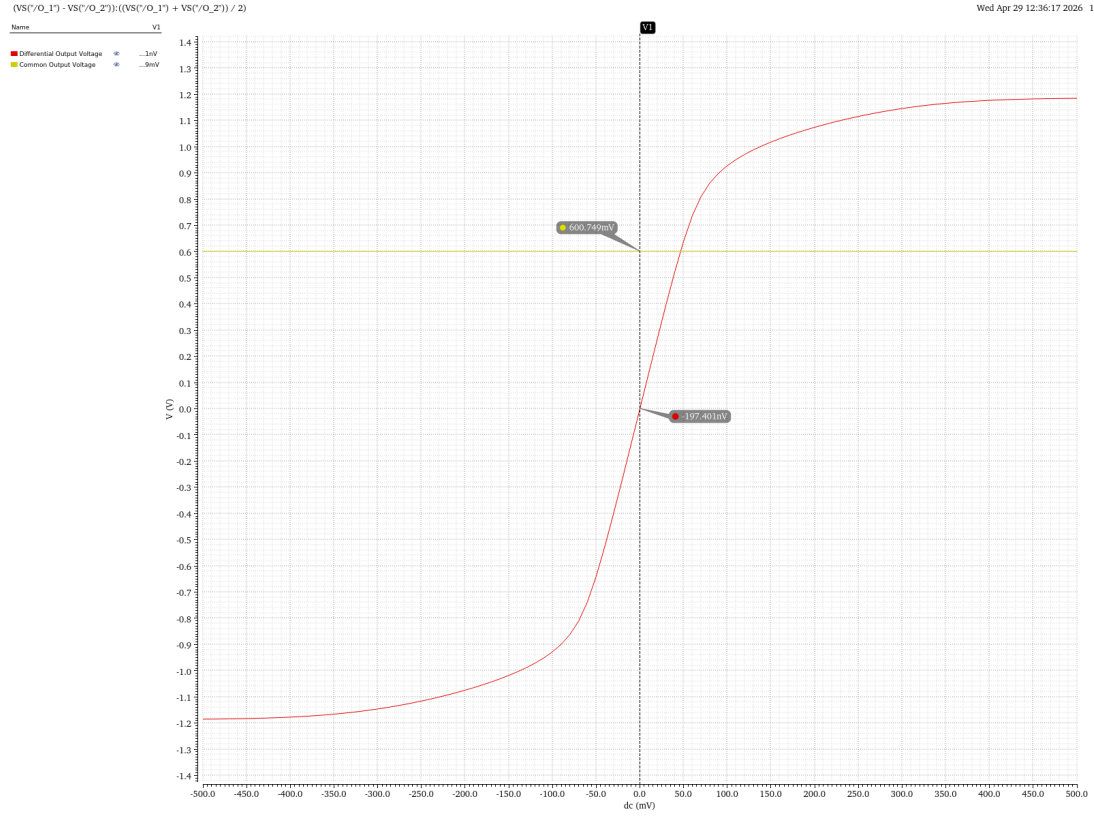


Figure 17: Differential and Common Output Voltage

From Fig. 18, the drain currents of the two transistors in the differential pair can be observed. As expected, I_{D1} and I_{D2} vary in a complementary manner: as V_{ID} increases, I_{D1} increases monotonically, while I_{D2} decreases monotonically. The nonlinear dependence of the drain currents on V_{ID} is also evident, in agreement with the theoretical behavior of a MOS differential pair.

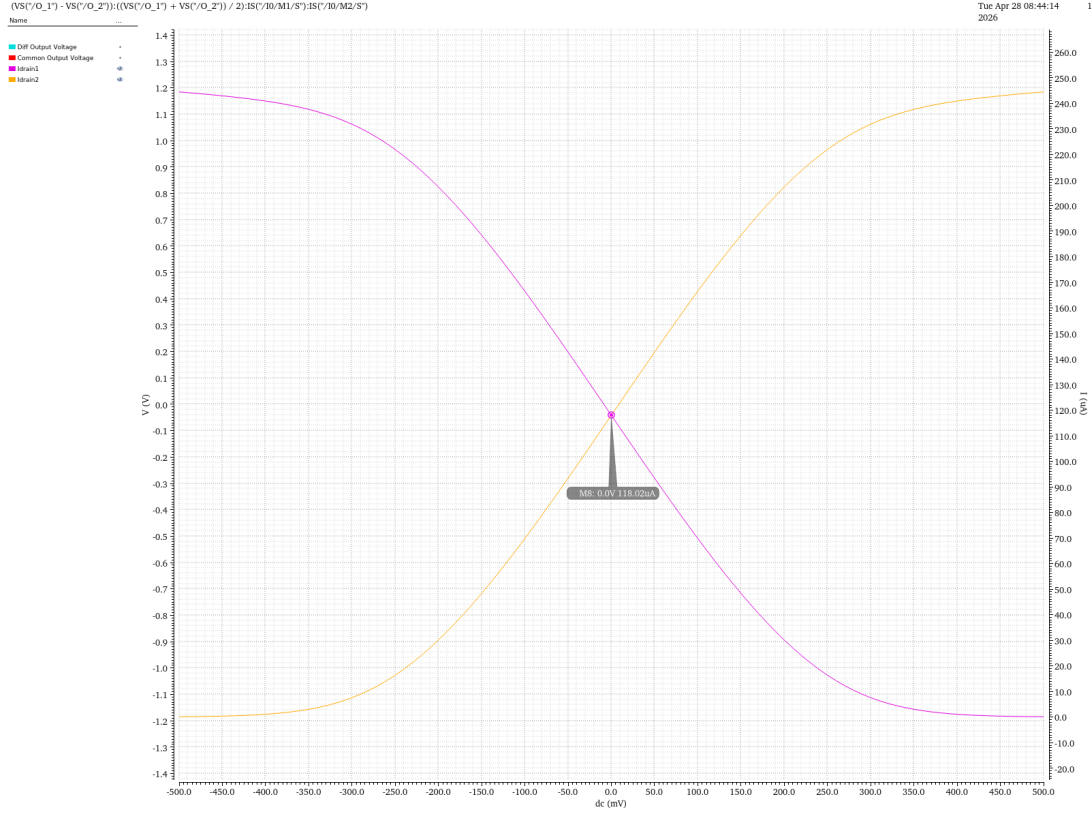


Figure 18: Drain Currents of the two transistor in the differential pair

An AC simulation was performed in order to evaluate the frequency response of the OTA. In particular, a Bode plot was generated to show both the magnitude and the phase of the differential-mode voltage transfer function.

From Fig. 19, the main small-signal frequency-domain parameters of the amplifier can be identified: the low-frequency gain A_{dd0} , the -3 dB bandwidth f_{-3dB} , and the unity-gain frequency f_u .

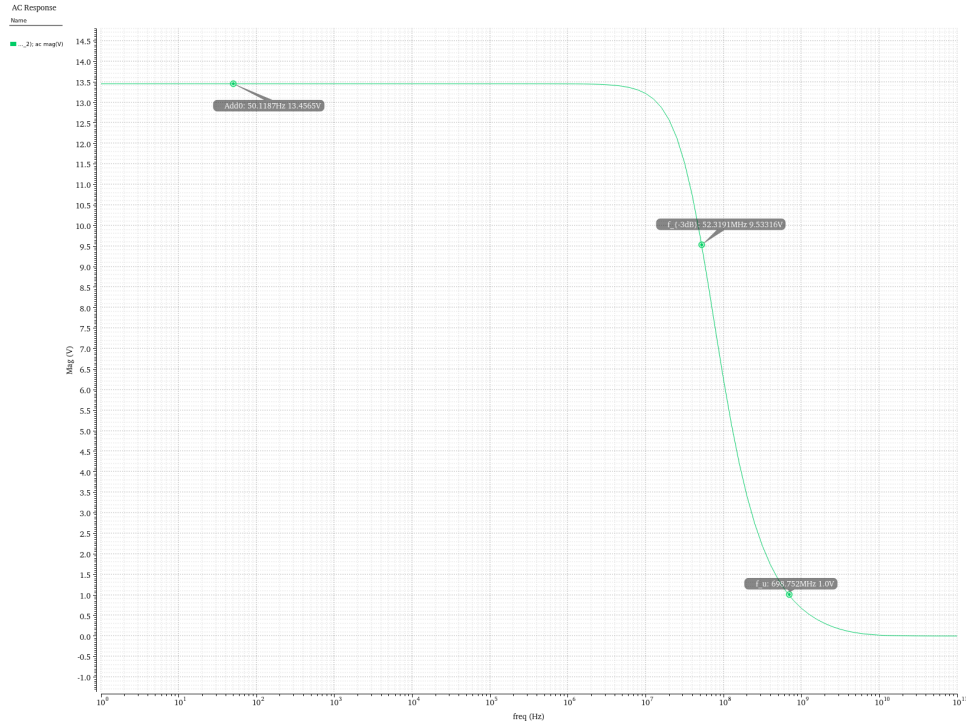
In the linear-scale plot of Fig. 19, the low-frequency gain is approximately $A_{dd0} = 13.4565$ V/V. The unity-gain frequency is found at the point where the magnitude crosses 1 V/V, which occurs approximately at 698.752 MHz. In the logarithmic plot of Fig. 10(b), the low-frequency gain corresponds to 22.5786 dB, while the -3 dB bandwidth is observed at approximately 52.1096 MHz, where the gain has dropped to approximately 19.5761 dB. The unity-gain frequency is approximately 694.105 MHz, where the magnitude intersects the 0 dB axis. The phase response is also shown in the same figure.

Since the -3 dB bandwidth is a good approximation of the dominant-pole frequency f_{dp} , the gain-bandwidth product can be estimated as

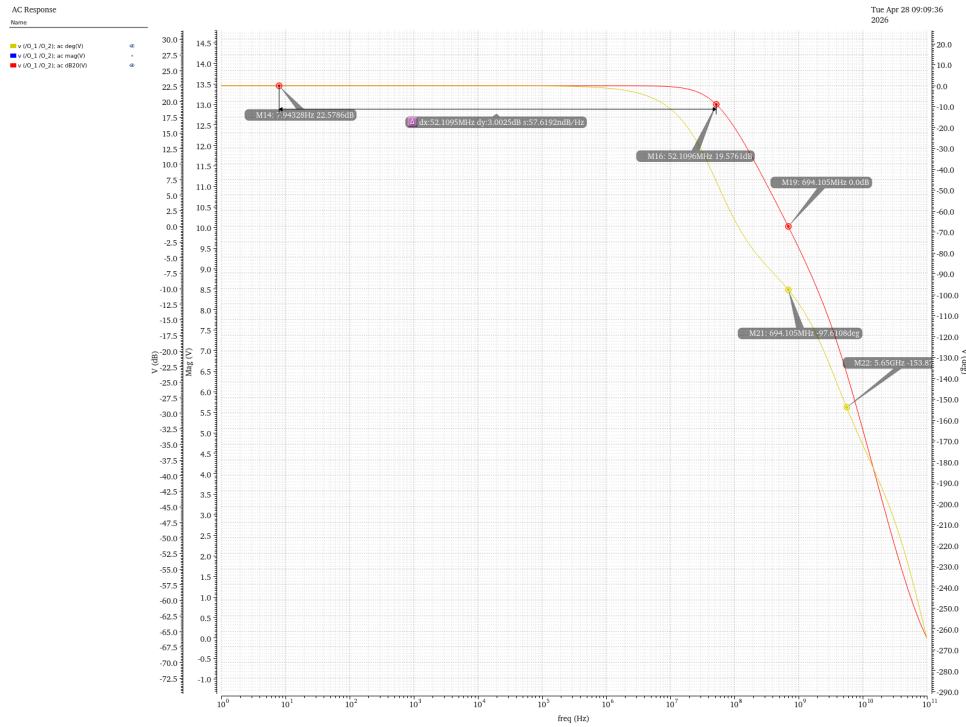
$$GBW \approx A_{dd0} f_{-3dB} \approx 13.4565 \times 52.1096 \text{ MHz} \approx 699.7 \text{ MHz}.$$

This value is very close to the unity-gain frequency extracted from the plot, which confirms that the amplifier is dominated by a single pole over the frequency range of interest. The second pole is therefore expected to be located much higher in frequency, on the order of 5–6 GHz, and thus well beyond f_u .

A second AC simulation was then performed by setting the AC component of the voltage sources V_{ID} and V_{IC} to 0 V and 1 V, respectively, in order to excite the OTA in



(a) Magnitude plot of the differential mode voltage transfer function - linear scale



(b) Magnitude plot of the differential mode voltage transfer function - logarithmic scale

Figure 19: AC Simulations of differential mode voltage transfer function

common-mode. Fig. 20 shows the corresponding Bode plot of the common-mode transfer function.

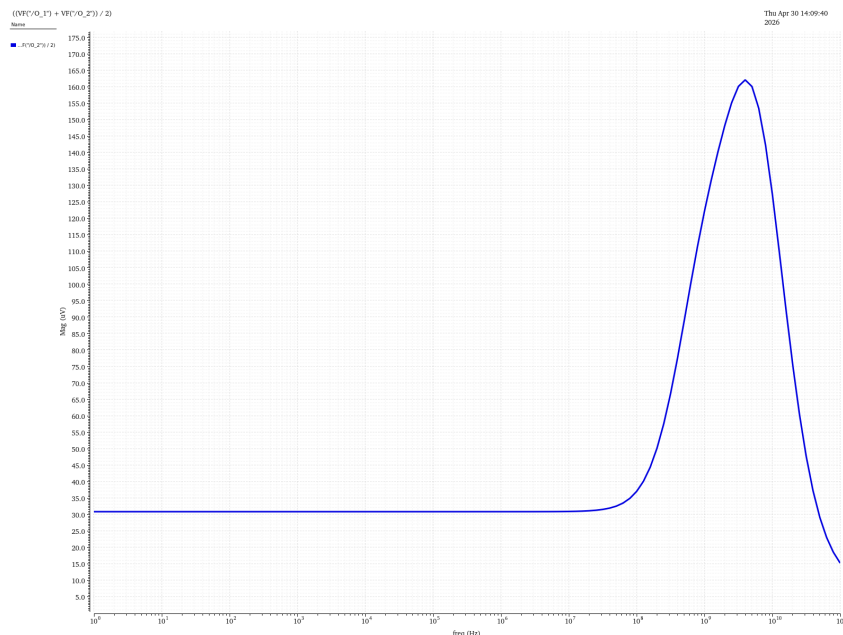


Figure 20: AC Simulation - Common Mode Gain

At low frequency, the common-mode gain is close to zero because the differential pair strongly rejects signals applied equally to both inputs. Since both transistors experience the same input variation, their drain currents change in nearly the same way and largely cancel at the differential output. The common-mode gain is not exactly zero mainly because of the finite output resistance of the tail current source and the presence of small device mismatches. This explains the very high low-frequency common-mode rejection ratio (CMRR).

2.3 Estimation of the OTA offset voltage V_{OS} with a Monte Carlo mismatch simulation

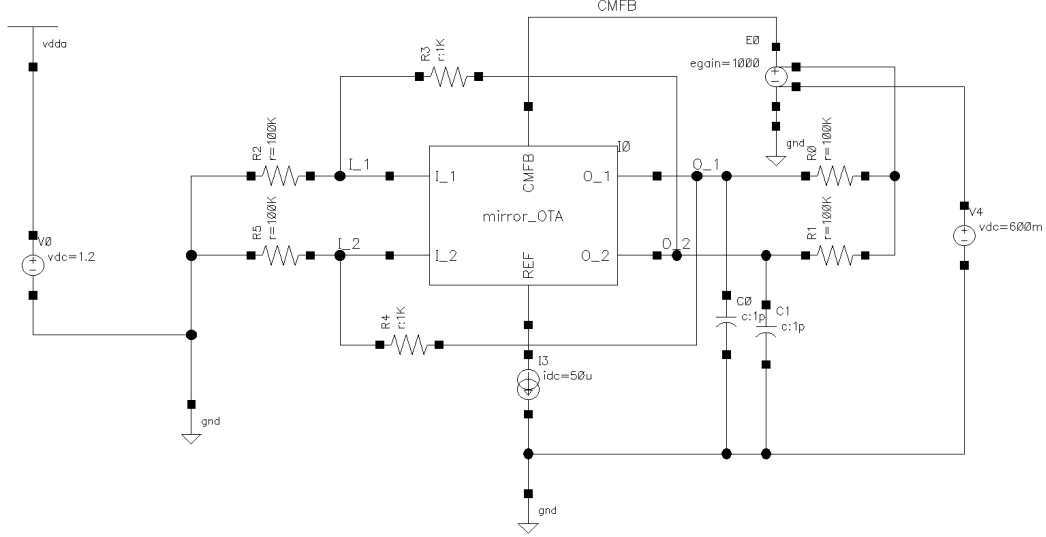


Figure 21: Test bench employed for the offset voltage evaluation.

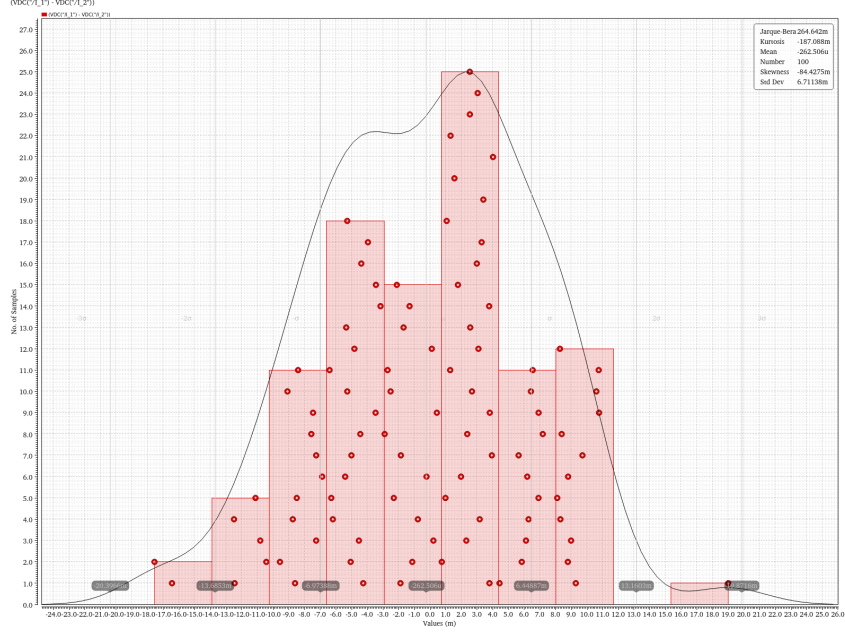
In order to extract the offset voltage we are going to run a Monte Carlo analysis with 100 different circuit samples. The new test bench used is reported in Fig. 21. From a theoretical estimation of the offset voltage considering only the process spread variation of transistors M_1 , M_2 , M_3 , M_4 :

$$\sigma_{\Delta V_{Tn}} = \frac{A_{\Delta V_{Tn}}}{\sqrt{W_3 \cdot L_3}} = 4.86 \text{ mV} \quad \sigma_{\frac{\Delta k'_n}{k'_n}} = \frac{A_{k'_n}}{\sqrt{W_3 \cdot L_3}} = 1.07\%$$

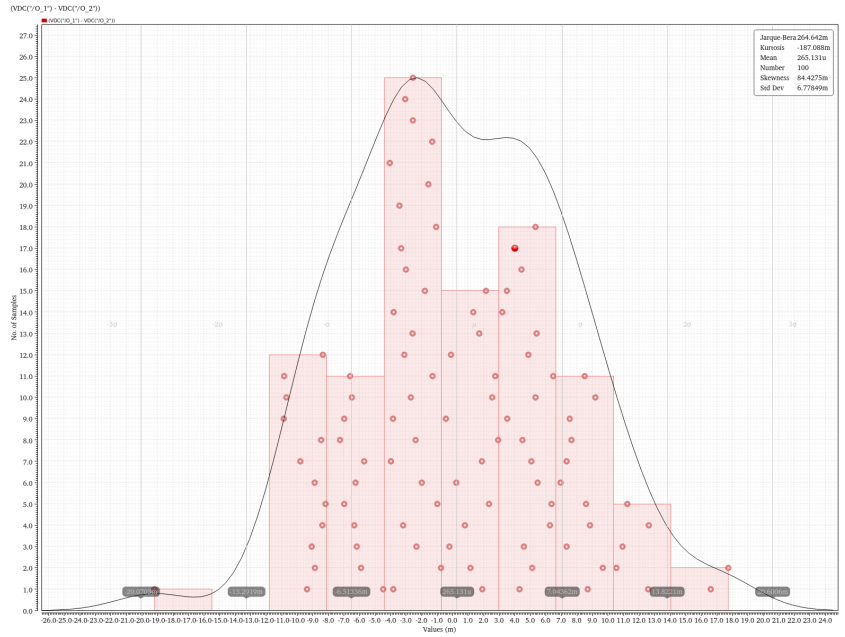
$$\sigma_{\Delta V_{Tp}} = \frac{A_{\Delta V_{Tp}}}{\sqrt{W_1 \cdot L_1}} = 3.77 \text{ mV} \quad \sigma_{\frac{\Delta k'_p}{k'_p}} = \frac{A_{k'_p}}{\sqrt{W_1 \cdot L_1}} = 1.047\%$$

$$V_{OS} = \sigma_{V_{ID}} = \sqrt{\left(\frac{V_{OV1}}{2}\right)^2 \left(\sigma_{\frac{\Delta k'_p}{k'_p}}^2 + \sigma_{\frac{\Delta k'_n}{k'_n}}^2\right) + \left(\frac{V_{OV1}}{V_{OV3}}\right)^2 \sigma_{\Delta V_{Tp}}^2 + \sigma_{\Delta V_{Tn}}^2} = 7.0 \text{ mV}$$

By computing the offset voltage V_{OS} , it is found that, with the current transistor parameters, the specification $V_{OS} \leq 4 \text{ mV}$ is not satisfied. Also, thanks to the Monte Carlo simulation, and thanks to the fact that the feedback factor is almost unitary, it is possible to say that the standard deviation in the simulation is also the offset voltage of



(a) Monte Carlo analysis of V_{id}



(b) Monte Carlo analysis of V_{od}

Figure 22: Monte Carlo Simulation

the output voltage differential mode. It is possible to see that even for the simulation, the specification is not satisfied. Therefore, the CM OTA must be designed and optimized so that all the required specifications are met.

2.4 Design of the CM OTA to Meet All Specifications

First of all, some parameters were fixed to help with the design :

$$GBW = 1.7 \text{ GHz} = 1.6 \cdot 10^9 \text{ Hz}$$

$$A_{dd0} = 30.5 \text{ dB}$$

From the circuit analysis, it is possible to say that :

$$GBW = A_{dd0} \cdot f_{p1} = \frac{G_m}{2\pi C_L} \implies G_m = GBW \cdot 2\pi C_L = 10 \text{ mS}$$

Once G_m was obtained, thanks to A_{dd0} , it is possible to evaluate R_o , the output resistance of the mirror OTA :

$$A_{dd0} = 10^{\frac{30.5}{20}} = G_m \cdot R_o \implies R_o = \frac{10^{\frac{35}{20}}}{G_m} = 3.349 \text{ k}\Omega$$

$$G_m = M \cdot g_{m1} = M \frac{2I_{d1}}{|V_{OV1}|} = M \frac{I_{B1}}{|V_{OV1}|} = \frac{2I_{d7}}{|V_{OV1}|}$$

Before continuing, it is necessary to choose the overdrive voltage of transistors M_1 and M_2 and the factor M , so $|V_{OV1}| = 0.125 \text{ V}$ and $M = W_5/W_3 = 5$ were set

$$\frac{I_{B1}}{2} = I_{d1} = \frac{g_{m1}|V_{OV1}|}{2} = 150 \text{ }\mu\text{A}$$

$$I_{B1} = 2 \cdot I_{d1} = 300 \text{ }\mu\text{A}$$

$$I_{B2} = M \cdot \frac{I_{B1}}{2} = 750 \text{ }\mu\text{A}$$

$$N = \frac{I_{B1}}{I_{ref}} = \frac{300 \text{ }\mu\text{A}}{50 \text{ }\mu\text{A}} = 6$$

Regarding transistors M_1 and M_2 , the gate length $L = 240 \text{ nm}$ was chosen and the gate width of the two transistors was evaluated, since $M_1 = M_2$:

$$W_2 = W_1 = L_1 \cdot \frac{2I_{d1}}{k'_p \cdot V_{OV1}^2} = 32 \text{ }\mu\text{m}$$

Thanks to the output resistance R_o found before, it was possible to find the gate length of transistor M_5 from the following expression :

$$R_o = r_{o5} || r_{o7} = \frac{\eta_n \eta_p}{\eta_n + \eta_p} \cdot \frac{L_5}{I_{d5}} \implies L_5 = \frac{R_o I_{d5}}{\eta} = 396 \text{ nm} \rightarrow 400 \text{ nm} = L_7 \quad \text{where } \eta = \frac{\eta_n \eta_p}{\eta_n + \eta_p}$$

By symmetry, $L_6 = L_8 = 400 \text{ nm}$. Since the couples M_3M_5 and M_4M_6 make a current mirror, $L_3 = L_4 = 400 \text{ nm}$ were obtained. At the end, for the tail current of M_9 which mirrors the reference current of M_{10} , the gate length has been chosen equal to the others: $L_9 = L_{10} = 400 \text{ nm}$. Now, from the specification on the output voltage swing we get:

$$V_{OV5} = |V_{OV7}| = 0.25 \text{ V}$$

Having the overdrive voltage of transistor M_5 and M_7 , and the drain current of all transistors, it is possible to evaluate the gate width for all transistors:

$$W_6 = W_5 = L_5 \cdot \frac{2I_{d5}}{k'_n V_{OV5}^2} = 20.4 \text{ } \mu\text{m} \rightarrow 20 \text{ } \mu\text{m}$$

$$W_4 = W_3 = \frac{W_5}{M} = \frac{W_5}{5} = 4.08 \text{ } \mu\text{m} \rightarrow 4 \text{ } \mu\text{m}$$

$$W_7 = W_8 = L_7 \cdot \frac{2I_{d7}}{k'_p V_{OV7}^2} = 96 \text{ } \mu\text{m}$$

$$\frac{W_7}{W_{10}} = \frac{M \cdot N}{2} = 15 \implies W_{10} = \frac{W_7}{20} = 6.4 \text{ } \mu\text{m}$$

$$\frac{W_9}{W_{10}} = N \implies W_9 = N \cdot W_{10} = 38.4 \text{ } \mu\text{m}$$

Lastly, the number of fingers and the width of these have to be chosen in order to have a perfectly matched current mirror. The gate width of the fingers $W_{p,finger}$ is equal to $W_{p,finger} = 3.2 \text{ } \mu\text{m}$, which is set to be equal to $\frac{W_{10}}{2}$. Thanks to this, it is possible to say the following :

$$M_9 : \frac{W_9}{W_{p,finger}} = \frac{38.4 \text{ } \mu\text{m}}{3.2 \text{ } \mu\text{m}} = 12$$

$$M_7 - M_8 : \frac{W_8}{W_{p,finger}} = \frac{96 \text{ } \mu\text{m}}{3.2 \text{ } \mu\text{m}} = 30$$

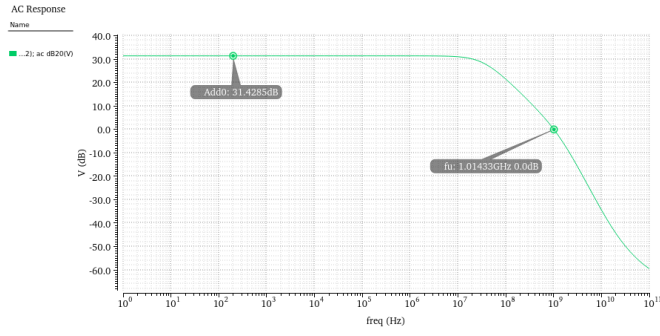
For the differential pair, the same gate width of the fingers is used as in $W_{p,finger}$, so :

$$M_1 - M_2 : \frac{W_1}{W_{p,finger}} = \frac{32 \text{ } \mu\text{m}}{3.2 \text{ } \mu\text{m}} = 10$$

However, for the n-type current mirror, the gate width of the fingers $W_{n,finger}$ is equal to $W_{n,finger} = 2 \text{ } \mu\text{m}$, which is set to be equal to $\frac{W_4}{2}$.

$$M_3 - M_4 : \frac{W_3}{W_{n,finger}} = \frac{4 \text{ } \mu\text{m}}{2 \text{ } \mu\text{m}} = 2$$

$$M_5 - M_6 : \frac{W_5}{W_{n,finger}} = \frac{20 \text{ } \mu\text{m}}{2 \text{ } \mu\text{m}} = 10$$



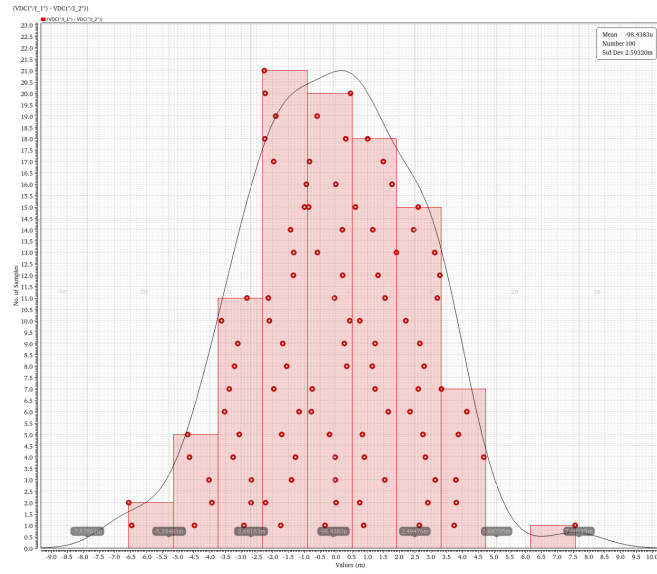
It is possible to notice from Fig. 14 and Fig. 15 that all specifications were met :

$$A_{dd0} = 31.4285 \text{ dB}$$

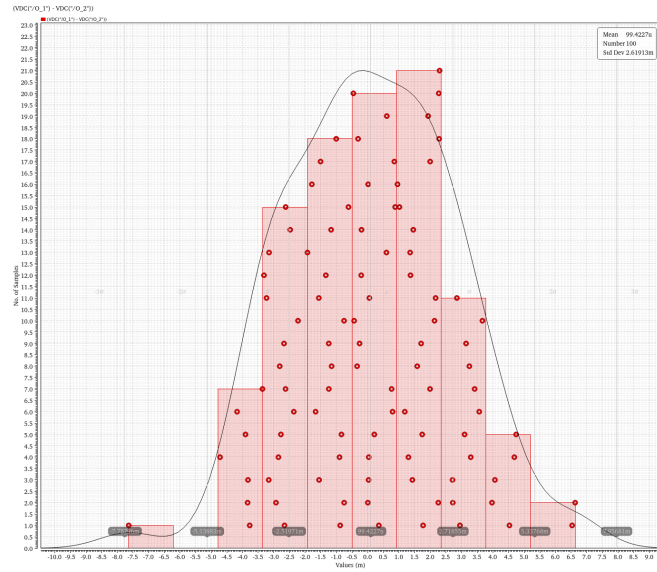
$$GBW = 1.014 \text{ GHz}$$

$$V_{OS} = 2.6\text{mV}$$

Figure 23: Differential mode voltage transfer function



(a) New Monte Carlo analysis of V_{id}



(b) New Monte Carlo analysis of V_{od}

Figure 24: Monte Carlo Simulation after the design of the CM OTA